

Right set of log files: Not every log that the application generates needs to be sent to cloud logging. Analyse and decide which logs are needed for analysis and troubleshooting purposes and filter out the unwanted ones while configuring the Ops Agent for integration.

Application logger configuration: Review the logger configuration of your application to ensure that the logs are at the right configuration level. As a best practice, production deployments should have the default logging at the info or warn level and not at the debug level, which should be enabled only temporarily for troubleshooting.

High-volume entries: Inspect the application logs in ‘Logs Explorer’ for unwanted, high-volume entries. Utilise the ‘Hide similar log entries’ feature to identify patterns and determine if they’re necessary.

Retention time: Logging, by default, comes with a 30-day retention period, with higher retention incurring additional costs. Thirty days is usually sufficient for most of the logs needed for application troubleshooting. Higher retention is needed only for special logs (e.g., logs stored for audit purposes). In such cases, ensure that such logs are routed separately to a different logging

bucket and the higher retention period is applied only for that bucket.

These are some of the ways in which we can ensure that the right set of logs is sent to cloud logging and the corresponding costs are at optimal levels.

Figure 2 gives a four-month cost report for an application where some of the cost optimisation strategies mentioned above were applied. In the ‘% change’ column, we can see the realised savings of approximately 20% in compute engine costs and around 88% in cloud logging costs (when compared to the previous four months).

Cloud storage

Cost-effective storage: Cloud storage costs per GB per month are relatively low, making it a cost-effective solution for regular applications that use such storage for data transfer or sharing. However, for data-intensive applications that retain TBs of data, costs can add up quickly.

Storage classes: To optimise costs, we can look at the different storage classes provided by cloud storage. ‘Coldline’ and ‘Archive’ storage classes offer significantly lower prices than the default ‘Standard’ storage. If there is data with infrequent access needs, it can be stored using low-cost storage options

like Coldline or Archive.

Automation: We can automate such cost saving actions using the ‘Object Lifecycle Management’ feature of cloud storage. We can create rules on the ‘Cloud Storage’ bucket – e.g., to downgrade storage class of objects older than 90 days to Coldline storage, delete objects older than 60 days, etc.

GCP Secret Manager

The GCP ‘Secret Manager’ service is used to securely store sensitive data. The costs associated with this service depend on the number of secrets stored and the number of access requests. In scenarios where the costs are high due to a huge volume of access requests, applications can consider caching such secrets in-memory, ensuring robust safeguards are in place to prevent any in-memory leaks. Such a caching approach will reduce the volume of access calls and the associated costs.

Multi-regional resiliency

GCP provides a very high uptime and great SLA metrics for their cloud services when deployed with a regional availability (single region with multiple zones) configuration. While this configuration suffices for most services,

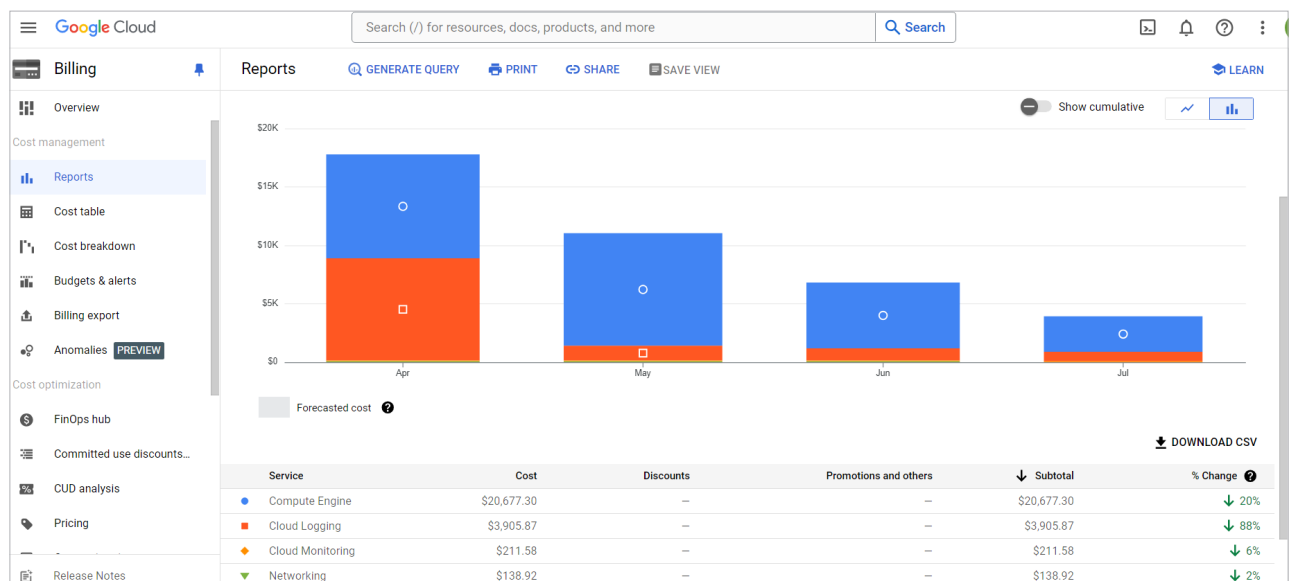


Figure 2: Reports showing reduced costs for an application after applying some of the mentioned strategies